HOUSE PRICE ANALYSIS

Exploratory Data Analysis, Data Cleaning & Feature Engineering

Dataset	02_house_price.csv
Original size	1,010 rows × 5 columns
Target variable	price_lakh
Analysis	Basic EDA + Feature Engineering

1. Problem Statement

Analyze house characteristics to understand which factors are associated with house price. The workflow identifies data-quality issues, explores relationships, cleans the dataset, engineers useful features, and separates predictors from the target.

2. Dataset Description

Column	Description
area_sqft	House area in square feet
bedrooms	Number of bedrooms
age_years	Age of the house in years
distance_city_km	Distance from city in kilometers
price_lakh	House price in lakh — target

3. Data Quality Findings

Check	Result
Original dataset	1,010 × 5
Duplicate rows	60
Total missing values	188
Invalid records	33
Price outliers using IQR	16

Invalid-value rules: negative area_sqft, bedrooms greater than 10, and negative price_lakh were treated as invalid.

4. Price Outlier Analysis

The Interquartile Range (IQR) method was applied to positive, non-missing prices. Values below Q1 − 1.5×IQR or above Q3 + 1.5×IQR were flagged as outliers. The analysis identified 16 price outliers.

5. Exploratory Visualizations

Visualization	Purpose
Area distribution	Inspect house-size spread
Price distribution	Inspect price spread and skewness
Area vs price	Study size–price relationship
Bedrooms vs price	Compare price by bedroom count
Age vs price	Study age–price relationship
Price boxplot	Check price outliers
Correlation heatmap	Summarize numerical correlations

6. Data Cleaning

1. Remove duplicate rows. 2. Replace invalid area, bedroom, and price values with NaN. 3. Median-impute remaining missing values column-wise. 4. Re-check the cleaned dataset.

Post-cleaning metric	Value
Final rows	950
Missing values	0
Duplicate rows	0
Negative area rows	0
Bedrooms > 10 rows	0
Negative price rows	0

7. Feature Engineering

Feature	Formula / Rule	Purpose
price_per_sqft	price_lakh / area_sqft	Normalizes price by house size
is_new	1 if age_years < 5, else 0	Identifies relatively new houses

8. Feature / Target Separation

After feature engineering, the dataset contains 950 rows and 7 columns. price_lakh is separated as the target variable.

Object	Shape	Contents
X	950 × 6	Predictor features
y	950	price_lakh target

9. Python Implementation

```
import pandas as pd
import numpy as np

df = pd.read_csv("02_house_price.csv")
clean = df.drop_duplicates().copy()
clean.loc[clean["area_sqft"] < 0, "area_sqft"] = np.nan
clean.loc[clean["bedrooms"] > 10, "bedrooms"] = np.nan
clean.loc[clean["price_lakh"] < 0, "price_lakh"] = np.nan

for col in clean.columns:
    clean[col] = clean[col].fillna(clean[col].median())

clean["price_per_sqft"] = (clean["price_lakh"] / clean["area_sqft"]).round(4)
clean["is_new"] = (clean["age_years"] < 5).astype(int)
X = clean.drop(columns=["price_lakh"])
y = clean["price_lakh"]
```

10. Final Results

Metric	Final value
Original dataset	1,010 × 5
Duplicates identified	60
Missing values identified	188
Invalid records identified	33
Price outliers identified	16
Cleaned dataset	950 × 5
After feature engineering	950 × 7
X	950 × 6
y	950

11. Conclusion

The dataset was systematically inspected and cleaned. Duplicate rows, missing values, and invalid records were handled, while price outliers were identified using IQR. Exploratory visualizations were prepared for distributions and relationships. Two useful features—price_per_sqft and is_new—were engineered, and price_lakh was separated as the target. The resulting dataset is ready for downstream statistical analysis or machine-learning modeling.

12. Learning Outcomes

• Inspect a numerical dataset and understand its structure and data types.
• Identify missing and duplicate records.
• Detect impossible values using domain rules.
• Apply IQR-based outlier detection.
• Visualize distributions and variable relationships.
• Clean data using duplicate removal and median imputation.
• Engineer meaningful features.
• Separate predictors X and target y.